

AI Customer Service Automation Workflow

Objective

Build an AI-powered automation workflow that starts automatically whenever a new order is added to the database.

The system's purpose is to automatically collect customer conversations, download artwork files, organize them, and prepare them for production with minimal human intervention.

Workflow

Step 1 – New Order Trigger

When a new order is inserted into the Order Database, automatically start the workflow.

The order contains information such as:

- Order ID
- Tracking Number
- Customer Name
- Phone Number
- Email
- Platform
- Ordered Products
- Quantity

Step 2 – Search Customer Conversations

The AI searches all communication channels using:

- Order ID
- Tracking Number
- Phone Number (if needed)

Communication channels:

- WhatsApp
- Email

- Shopee Chat
- Lazada Chat

Every conversation, message, and attachment found must also be synchronized and saved into the Chat Database.

Step 3A – Conversation Not Found

If no conversation or attachment is found:

- Create a "Waiting for Customer Files" task.
- Automatically remind the customer to send the required artwork.
- Retry searching after each customer reply.

If the customer still does not send the files after the configured reminder limit, move the case to Human Review for manual handling.

Step 3B – Conversation Found

If the conversation is found:

- Download all customer attachments.
- Save them to the Local Network Storage.
- Link every downloaded file to the corresponding order.

Step 4 – AI File Processing

The AI analyzes the order and determines which file belongs to each ordered product.

Then:

- Match each file to the correct product.
- Rename every file using the product name and order information.
- Organize the files inside the order folder.

Example:

12345_Sticker_A5.pdf

12345_Business_Card.ai

12345_Banner.jpg

Databases

Order Database

- Stores:

Orders

Order Items

- Customer Information

This database triggers the workflow.

Chat Database

Store all conversations, including:

- WhatsApp messages
- Email messages
- Shopee Chat
- Lazada Chat

For every message, save:

- Platform

Conversation ID

Message ID

- Sender

Receiver

Message Content

Message Time

- Order ID (if detected)
- Tracking Number (if detected)

Attachments

Attachment Location

The Chat Database becomes the central repository for all customer communication.

File Storage

Store all downloaded customer files in the Local Network Storage after they have been renamed and matched to the correct product.

Human Review

- Cases should be sent to Human Review when:

No conversation is found.

No attachment is found.

AI cannot confidently match files to products.

Multiple possible conversations are found.

Manual customer communication is required.

Roles

Order Database

- Detect new orders.
- Trigger the workflow.

Chat Integration

- Synchronize conversations from all communication platforms.

Chat Database

Store every conversation, message, and attachment.

AI Search Agent

- Search conversations.
- Match orders with customers.
- Detect attachments.

AI File Agent

- Download files.
- Match files to ordered products.
- Rename files.
- Organize files.

Automation Engine

- Control the workflow.
- Send reminders.
- Retry searches.
- Update workflow status.

- Move failed cases to Human Review.

Local Network Storage

Store renamed customer files.

Human Review

- Handle unresolved cases.
- Contact customers manually.
- Approve or correct AI decisions.

Goal

- The entire workflow should run automatically after a new order is created.
- The only time a human should be involved is when the AI or automation cannot confidently complete the process. The system should maintain complete records of orders, conversations, messages, attachments, workflow status, and file locations to ensure full traceability and efficient production preparation.